

Institute of Actuaries of India

Subject CP3 – Communication Practice

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Re: Framework for assessing performance of investment managers

Dear Geeta,

Thank you for your mail. My team has prepared a note with calculations for some schemes and my recommendation for the framework is based on this note.

In suggesting a framework I have considered three important factors which I have listed below.

- The calculation of the return itself
- How this return compares with the return the customer would have earned by investing directly in the appropriate stock market index
- Whether the return earned is sufficient considering the risks the fund managers would have assumed.

I have explained this in further detail in the following paragraphs.

The calculation of investment returns is straightforward when there are no flows into and out of the fund during the period of evaluation. However, these being pension schemes, there would be contributions into the fund as the employer funds the pension liabilities or there could be withdrawals as members exit the pension scheme. The timing of these cash flows would have an impact on the return achieved and therefore the calculation of investment returns should appropriately allow for the effect of these inflows and outflows.

The two measures which we are considering are the “Time weighted rate of return” (Twrr) and the “Money weighted rate of return” (Mwrr). Both the measures are correct in their respective contexts. However, we need to arrive at the right measure in the context of the evaluation of the performance of the fund managers.

The Mwrr gives weightage to the returns earned when the funds were higher. To explain this further if a large contribution is made into a scheme and immediately thereafter the market rallies and stays high till the end of the evaluation period, the return calculated using this measure for the scheme which had the inflow would be higher as compared to a scheme which did not have such a contribution. The return thus calculated is not very informative of the skill of the fund manager. Similarly if the market falls after such a contribution, then the return would be lower. The fund managers do not control the inflows and outflows from the schemes and therefore we need to isolate the effects of these flows. We therefore calculate the Twrr which is an average return over the period under consideration. This return which is not biased by cash flows is more indicative of the skill of the fund manager. This measure should form the primary basis for evaluation of performance of the fund manager.

While the scheme inflows and outflows cannot be controlled by the fund managers in reality the scheme trustees do interact with the fund managers when making large contributions or withdrawals from the schemes. In addition, the pension scheme trustees measure returns and compare them with the returns which has been used in calculating the pension scheme liabilities. We should therefore allocate some weight to this measure in evaluating the performance.

While we have discussed the measures of returns, we do need to recognize that the fund performance has to be compared to the returns that could have been earned by an investor investing directly in the index or the benchmark. When the scheme trustees entrust the funds to us they do expect that we deliver higher returns than what they would have earned by investing in say an index fund. Therefore a fund manager who has delivered a relatively higher return as compared to another fund manager is the better one. In assessing this excess we should consider the Twrr as we said this measure is the one which reflects the skill of the fund manager.

There is one complication in looking at this absolute excess return. We need to see if this excess return has been achieved by taking higher risks. Higher risk means that the return could vary significantly over time. In order to allow for the risks

assumed by the fund managers in measuring performance, we calculate a “reward to risk ratio” which is the Sharpe ratio. The higher this ratio, the better the performance of the fund manager as the returns earned have been earned by not taking excessive risks or the returns earned are commensurate with the risks assumed.

I have summarized below for two of the large schemes the calculations done on the above basis for more clarity.

	Fund A	Fund B
Twrr	9.3%	8.7%
Mwrr	8.6%	11.6%
Return in excess of benchmark	3.8%	3.2%
Reward to risk ratio (Sharpe ratio)	1.3	1.4

We can see in this scenario that Fund manager A has performed better than Fund manager B on Twrr basis. Mwrr is higher for Fund B as for this particular scheme markets fell after a large withdrawal. Both funds have given positive returns over the benchmark. However, when reward to risk ratio is considered Fund B has the better performance as it has been achieved with lesser risk.

My recommendation for assessing performance of investment managers is to accord 50% weight to returns in excess of benchmark returns calculated on Twrr basis as this is reflective of their skills. We would also like our fund managers to not take excessive risks in order to earn higher returns and therefore the risk adjusted return ratio should have a weightage of 40%. However clients compare Mwrr against their expected returns and hence this can have a nominal weightage of 10%.

I trust this is clear and I would be happy to answer further questions if any.

Regards,
Lata,

[90 Marks]

Solution 2:

Any point relevant what the candidate describes as being appropriate.

One possible answer

I ensured that the tone of the email was appropriate by considering that she would have sufficient knowledge of pension schemes and of stock markets and indices. I did not explain these terms and have used these terms without putting in explanations and ensured that the mail appeared not to “talk down”. At the same time, I did not think there was a need to explain the calculations as her focus would be in understanding that the framework is relevant rather than the actual calculations and the associated complications of the calculations.

[2 Marks]

Solution 3:

Anything relevant which the candidate summarises to facilitate better understanding of the framework and recommendations

One possible answer

The paper produced by the investment team had a lot of data which had been used to arrive at the key results. The data which was required to actually perform the calculations did not have to be provided here again. I just ensured a summary was provided of the two measures so that it was clear that the returns could vary based on the measure adopted. The

summarising of the data also highlighted the issues that were involved in deciding relative performance of investment managers

[4 Marks]

Solution 4:

Any of the terms which the candidate explains

One possible answer

I explained the concept of TWRR and MWRR. While I named the Sharpe ratio I did not do into any further explanations as I did not think the Head of HR would be interested in the actual calculation.

[4 Marks]
